

Recoverable but unrealized: the mask-quality gap of prototype-based instance segmentation

Target journal: Pattern Recognition Letters.

Document status: Complete scientific draft for author review; not submitted.

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Author-supplied declarations

Funding and grant identifiers: _____

Competing interests: _____

CRediT contributions by author: _____

Data/code release destination: _____

Known data and implementation sources

The study uses COCO val2017 annotations and images (5,000 images, 36,335 ordinary GT), the COCO-pretrained Ultralytics YOLO26m-seg weights, and Ultralytics 8.4.100. All readouts are diagnostics on the frozen checkpoint; no weights are trained. Every number in the manuscript traces to a retained run record in the project archive (protocols frozen before execution, per-candidate/per-GT records, independent verification runs, source snapshots with hashes, transfer manifests); the claim-to-record map is maintained as the study evidence map. Dataset images and segmenter weights are not bundled.

Generative AI assistance recorded for this draft

OpenAI Codex (GPT-6) assisted research-protocol development, Python implementation, literature checking, statistical verification, manuscript organization and drafting, and table construction. All numbers were produced by executed, hash-verified runs of the project's experiment pipeline; no AI-generated measurements are used. The final authors must review the text and code, specify their contributions, and complete their own responsibility declaration before submission. This document does not assert that this review has already occurred.